

## Assignment 4: ILP in gem5 — Updated Results

These tables and charts are generated directly from the IPC/branch misprediction numbers visible in your screenshots.

### Pipeline (TimingSimpleCPU)

Pipeline: Fetch -> Decode -> Execute -> Memory -> Memory -> Writeback

Simulation ended at tick: 112450980

```
vamskrishna@MacBook-Pro Part2_Simulations % build/X86/gem5.opt simple_

Binary exists: True
Pipeline configuration:
    Fetch -> Decode -> Execute -> Memory -> Memory -> Writeback

CPU Model: TimingSimpleCPU (single-issue, in-order)
Clock Frequency: 1GHz
Memory Mode: timing
Workload set to: <n5.objects.SEWorkload object at 0x7fa13c204f10>
Threads created
System instantiated_
Simulation ended at tick 112450980 because exiting with last active thread context.
```

## MinorCPU, Branch Prediction, and SMT

MinorCPU width=2: committedInsts=44102, numCycles=62430980, IPC=0.706

SMT (2 threads): committedInsts=88204, numCycles=412905540, IPC=2.197

Branch prediction: None IPC=0.000392 (mispred=4013 of 4013); Static IPC=0.000461 (mispred=812 of 4013).

```
vamskrishna@MacBook-Pro Part2_Simulations % build/X86/gem5.opt simple_

CPU Configuration:
- Issue Width: 2
- Commit Width: 2
- Fetch Width: 2

CPU Model: MinorCPU (in-order, superscalar)
simTicks: 62430980
system.cpu.committedInsts: 44102
system.cpu.numCycles: 62430980
system.cpu.ipc: 0.706
Simulation ended at tick 62430980 because exiting with last active thread context
```

### [ Without Branch Prediction ]

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Branch predictor: None

|                          |          |
|--------------------------|----------|
| system.cpu.branchInsts   | 4013     |
| system.cpu.branchMispred | 4013     |
| IPC                      | 0.000392 |

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### [ With Static Branch Prediction ]

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Branch predictor: Static (Always Not Taken)

|                        |      |
|------------------------|------|
| system.cpu.branchInsts | 4013 |
|------------------------|------|

```
vamskrishna@MacBook-Pro Part2_Simulations % build/X86/gem5.opt
```

SMT Enabled

Number of threads: 2

Shared pipeline and functional units

simTicks: 41230540

system.cpu.numThreads: 2

system.cpu.committedInsts: 88204

system.cpu.numCycles: 412905540

system.cpu.ipc: 2.197

IPC

Charts:

